

Fertility and Housing Tenure Choice

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Question. How does access to family-sized housing affect births, and which housing policies can change fertility?

- Mechanism tests separate mortgage access, unsecured liquidity, and space costs.
- Revised household preferences and accounting support a new 2007 reference.
- Rebuilt housing evidence and richer fertility moments sharpen the fit diagnosis.
- Faster stationary computation makes controlled recalibration more practical.

The main remaining fit problem is the number of children among young mothers.

Source: PSID, CPS, AHS, and model experiments, September 17–30

Mortgage access and unsecured liquidity

Early diagnostics hold the child-benefit level and other parameters fixed:

	Baseline	No down payment	Unsecured credit
Completed fertility	1.872	1.866	1.948
Mean first-birth age	26.960	26.970	25.790
Ownership, ages 30–55 (%)	45.9	55.7	45.0

The unsecured-credit diagnostic allows five years of earnings.

- Removing the down payment primarily increases homeownership.
- Unsecured liquidity increases births and brings them forward.
- Family-sized rental space limits the connection between buying and births.

These are comparisons within the early sandbox, before the later specification and recalibration.

Household preferences and demographic accounting

A first-child housing-share loading replaces the hard housing floor:

$$\alpha(m) = \alpha_0 - \Delta\alpha \mathbf{1}\{m > 0\}, \quad v(m) = \psi m^{1-\gamma}, \quad v(0) = 0.$$

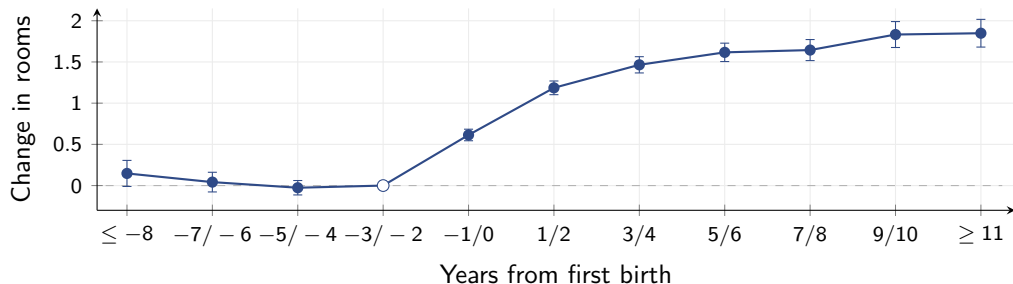
Here m counts children at home. $\alpha(m)$ is the consumption share, ψ the direct benefit level, and γ its curvature.

- Material utility retains CRRA, equivalence scales, and persistent earnings risk.
- Shares are compensated at a fixed rent, with no later-child loading.
- Dependency ends with probability 2/9 per four-year period. Adult entry is a separate queue, split equally after 16 and 20 years.
- Estates fund positive entrant assets. An explicit sink absorbs the remainder.

Concurrent children and lifetime births are distinct calibration objects.

Housing around the first birth

PSID adults who were the reference person or spouse before the birth:



At years 3–4: rooms rise by **1.465** (SE 0.050), ownership by **26.886 pp**, and recent moving falls by **21.147 pp**.

Rebuilt interview dates and baseline household status resolve earlier measurement differences. Whiskers show 95% pointwise intervals.

Source: PSID, Sun–Abraham event study, person/year effects, IW weights, September 24

Calibration inputs and empirical counterparts

The single housing market uses national inputs and a replacement-fertility reference.

Input	Value	Empirical source
Mean occupied rooms	5.729	AHS 2007
Pension / working-household earnings	0.229	CPS ASEC, income 2007
Annual child-directed estates / wealth	0.729%	SCF 2007 mortality proxy
Annual depreciation / property tax	1.416% / 1.060%	National housing inputs

- Baseline payroll tax is 8.028%. Equal retiree pensions balance PAYGO.
- The 1.465-room first-birth target has a model proxy, rather than a matched event-study estimator.
- Bequest and wealth targets retain recipient and income-definition gaps.

Fertility 2.1 normalizes the reference, rather than observed 2007 period births.

2007 stationary reference fit

Moment	Target	Model
Completed fertility (normalization)	2.100	2.100
Childlessness, ages 40–44 (%)	19.828	20.106
Exactly one child among mothers, ages 40–44 (%)	21.366	20.942
Mean first-birth age (years)	25.976	25.933
First births at age 30+ (%) [†]	24.928	22.365
Wealth / earnings	6.927	6.326
Annual child-directed bequest flow / wealth (%)	0.729	0.706
Older wealth/income p90/p50 [†]	3.516	3.069
Mean occupied rooms	5.729	5.848
Ownership, ages 30–55 (%)	67.626	65.482
First-birth rooms difference	1.465	1.622
Family rooms difference [†]	0.385	0.353
Recent-parent ownership difference (pp)	12.761	11.955
Children per woman at age 25, capped at three	0.810	0.535

[†]Untargeted validation. Ten scored moments and a separate fertility normalization.

Source: CPS, NCHS, PSID, AHS, SCF, and September 28 verified model export

Young mothers and the fertility intensive margin

Two unadopted experimental calibrations, with completed fertility normalized to 2.1:

Age-25 moment	Data	One birth per period	Two-birth option
Children per woman, capped at three	0.810	0.530	0.606
Mothers (%)	45.725	44.782	43.773
Children per mother, capped at three	1.770	1.185	1.385

- The motherhood share is close. Additional births among young mothers drive the miss.
- The two-birth option closes 27.1% of the early-count gap, while other fits deteriorate.
- Child-benefit curvature dominates a weak local parameter direction. A selected-point robustness check remains open.

The extra birth opportunity also adds an independent taste draw. The comparison does not isolate birth spacing alone.

House prices, credit, and births

Local elasticities from prescribed house-price and rent changes of $\pm 1\%$:

Outcome	Reference credit	Lifetime repayment only
Immediate total births	-0.437	-0.469
Immediate first births	-0.880	-0.906
Immediate second births	-0.114	-0.116
Completed fertility, recomputed cohort	-0.536	-0.555

First births account for 87.57% of the immediate decline under reference credit.

- Relaxing borrowing limits raises fertility levels at the common reference price.
- It leaves the local price response at least as strong in this comparison.

Preferences and entry stay fixed. These are prescribed-price responses, with lifetime solvency, rather than cleared equilibria or transition estimates.

Stationary computation and grid resolution

Matched comparisons preserve each experiment's economics and acceptance checks:

Comparison	Original	Revised	Result
Saving routine, same 160×15 grid	549 s	417 s	24.04% less time
Grid, 160×15 versus 120×9	469 s	187 s	60.04% less time

- The saving optimization reproduces policies, full moment tables, and plots exactly.
- Grid test with renter debt capped at 53% of mean annual earnings: ownership +0.201 pp, prices +0.076%.
- Ownership gaps above 5 pp cover 3.152% of mass in the interpolated comparison.

Local thresholds and transitions still require validation. The grid comparison also approximates the joint wealth–income entry distribution.

Entry wealth, credit, and the next calibration

Full repayment on sale into renting, renter saving floor $b' \geq -D$, and a 2% annual real rate:

Authorized one-hour exploratory arm	D (mean annual earnings)
Empirical wealth/earnings ratios, independent of income	0.25
Zero entrant wealth	0
Nonnegative ratio nodes, rescaled to preserve the mean	0

- Five-ratio diagnostic at $D = 0.53$: prices +0.089%, ownership -0.017 pp, with parameters fixed.
- Birth-renewal price clearing makes supply scale a population shifter. The new search fixes supply scale and child benefit.
- Historical-shock fits yielded no estimates. The pension diagnostic was cancelled.

Pilots submitted at 17:01 New York on September 30 and running at 17:26. Discussion: credit/entry, curvature restrictions, and matched empirical moments.